1. Regularization Overview:

techniques intended to improve generalization specifically by reducing the gap between training error and test error or avoiding overfitting, are collectively known as regularization techniques.

2. Regularization Technique — Early Stopping:

early stopping is a regularization technique that halts training when test error begins to increase.
it is effective only if the test error demonstrates a distinct U-shaped curve where the test error decreases and subsequently starts increasing.
if the test error remains flat or stagnant without rising, alternative regularization techniques must be used.

3. Regularization Technique — Weight Decay:

A. weight decay is implemented by appending a penalty term directly to the loss function.
this extra penalty forces the learning algorithm to drive down the magnitude of weights during training.
the weights contribute only to fitting a few specific training data are penalized and diminished, as they don't improve generalized capacity.
typically the bias weights are excluded from weight decay regularization.

B. L1 weight decay: loss function = original loss function + $\lambda \cdot \sum_{i=0}^{n} |w_i|$, [λ: a constant regularization hyperparameter that indicates strength of the penalty applied to a model.
w_i: the value of i-th weight
n: the total number of regularized weights across the corresponding layer in a neural network.]

C. L2 weight decay: loss function = original loss function + $\lambda \cdot \sum_{i=0}^{n} w_i^2$, [λ: a constant regularization hyperparameter that indicates strength of the penalty applied to a model.
w_i: the value of i-th weight
n: the total number of regularized weights across the corresponding layer in a neural network.]

4. Regularization Technique — Inverted Dropout:

A. during each epoch, a random fraction but specific proportion (= dropout rate) of neurons is temporarily zeroed out (= discard).
this prevents the certain group of perceptrons from co-adapting to memorize specific patterns and forces individual neurons to learn robust and generalizable features alongside arbitrary peer neurons.

B. crucially, inverted dropout scales the values of activation functions in all retained neurons by dividing them with keep probability (= 1 - dropout rate) directly during the forward pass.
because the values of activation functions in all retained neurons are compensated during the forward pass in the training stage.
the inference stage requires no additional weight scaling. all neurons operate at full capacity, simplifying and speeding up the inference process.

C.

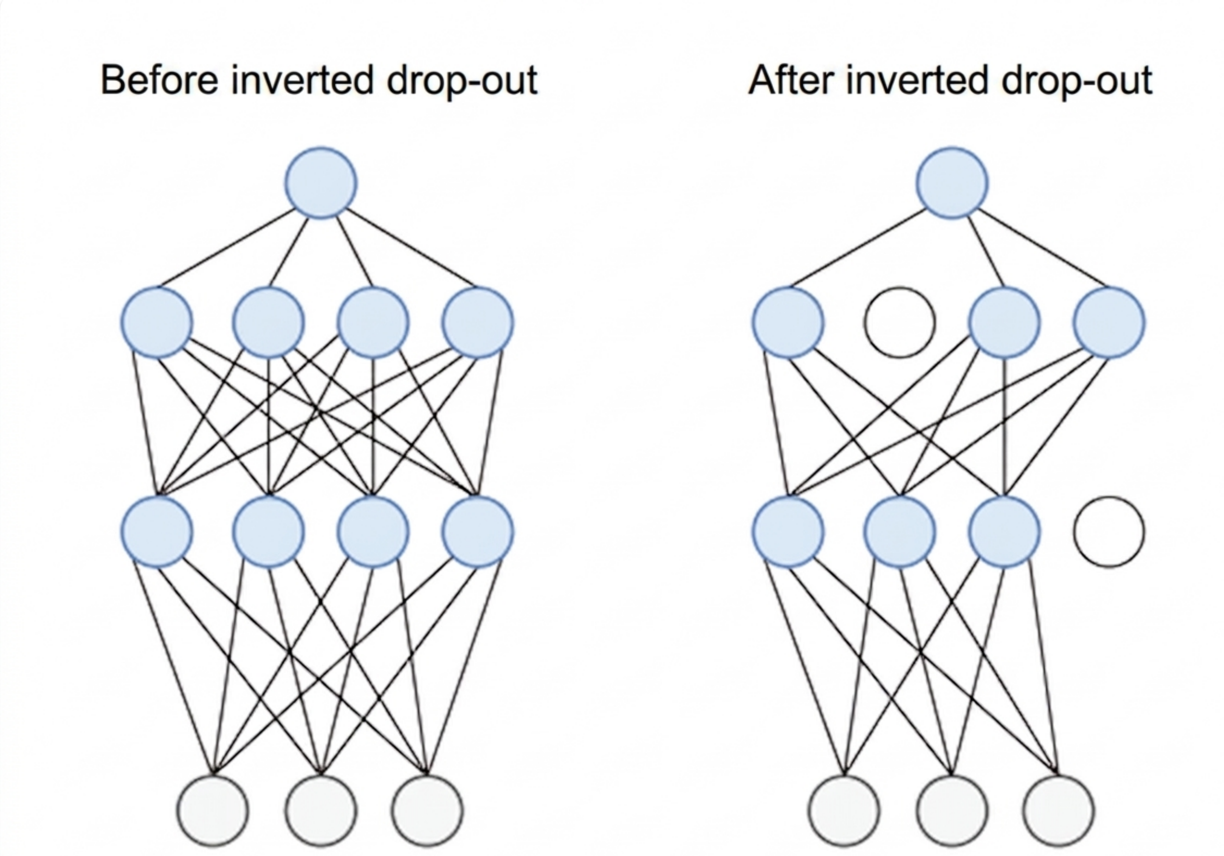

Figure 6-5 Inverted Dropout